State Feedback Control of a Continuous Steel Casting Process

FINAL PROJECT REPORT

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1. Context of the Problem

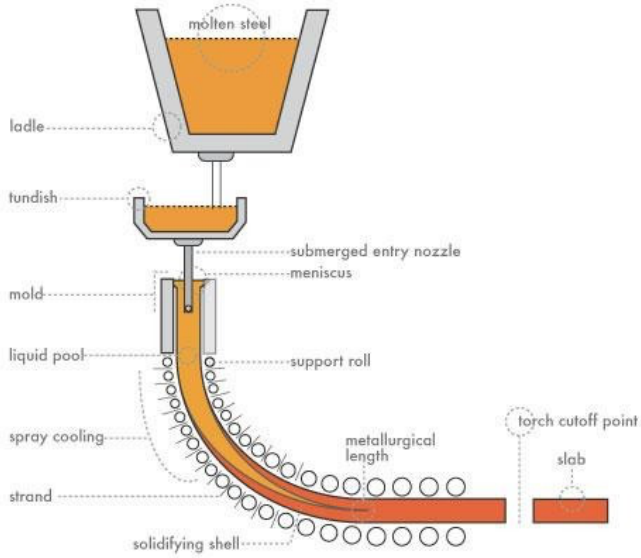

Figure 1.0: Continuous Steel Casting Process (Metallon Corporation, n.d.)

Steelmaking is a core pillar of the modern economy, and it poses numerous control challenges of academic and industrial interest. An important step in many steelmaking processes is **continuous casting**, outlined in Figure 1.0. Molten steel is transferred from a ladle to a mold via an intermediate container called a tundish. The metal is moved out of the mold and spray cooled as it travels along a line of support rollers. As it moves, it solidifies from molten metal into solid steel. The result is a "semi-finished" slab of steel that may be further processed into beams, rails, or other components.

This process must be carefully controlled to avoid defects and safety hazards. The **mold height** (the height of molten steel in the mold) is the primary control target for our project. If it becomes too high, molten steel can overflow the mold, creating a catastrophic safety hazard. If it is too low, air entrainment can occur, introducing defects into the steel and weakening it. Additionally, large oscillations in mold height can cause surface defects, forcing the steel to be scrapped. Many of the tools from EC402 find applications in the control of mold height, including state feedback, linearization, and state space modeling.

Molten steel levels are seldom perfectly observable in practical continuous casting systems. "Extreme temperatures, harsh conditions, slag interference, and continuous motion make molten metal measurement one of the toughest challenges in industrial automation" (Ripik AI, n.d.). Thus, continuous casting poses a natural extension of the EC402 course material by enabling exploration of the challenges posed by imperfect state observability and the design of observers.

Steelmaking is simultaneously a cost-sensitive and quality-sensitive industry. Cost and quality are in many instances anticorrelated, so steelmakers must navigate tradeoffs between them in a principled fashion. As an extension to our project, therefore, we will employ Linear Quadratic Regulator (LQR) control to compute feedback laws that optimally balance regulation performance and control effort. In this context, deviations of the mold level from its desired operating point correspond to quality losses and increased defect risk, while aggressive control actions correspond to actuator wear, energy consumption, and operational stress. The LQR framework provides a systematic way to trade off these competing objectives, yielding control policies that are both high-performance and operationally efficient.

2. Problem Statement

This project investigates the regulation of mold height in a continuous casting system using state-space control methods in the presence of disturbances.

Specifically, we:

- Develop a nonlinear state-space model of the continuous casting process from physical principles
- Linearize the model about a physically meaningful operating point

- Design and compare state feedback controllers using pole placement and Linear Quadratic Regulator (LQR) methods
- Extend the framework to the partial observability case using a Luenberger observer

Controller performance is evaluated through numerical simulation using metrics that reflect operational priorities, including oscillation intensity, deviation from the operating point, settling time, control effort, and violations of safety and quality constraints.

3. Methods and Analysis

3.1 Definitions and Model Construction

To formulate a state-space model for this control problem, we begin by defining the state vector

$$\boldsymbol{x}(\boldsymbol{t}) := \begin{pmatrix} h_M(t) \\ h_T(t) \end{pmatrix}$$

where h_M and h_T are the **mold height** and the **tundish height** (i.e., the height of molten steel in the mold and tundish) at time t, respectively. These states are coupled, as will be shown shortly.

We then define the input vector:

$$\boldsymbol{u}(\boldsymbol{t}) := \begin{pmatrix} u_M(t) \\ v_w(t) \end{pmatrix}$$

where $u_{M(t)} \in [0,1]$ is a flow regulation parameter representing the action of a slide gate or stopper rod modulating flow between the tundish and the mold and $v_w(t)$ is the **withdrawal speed**, the rate at which steel moves out of the bottom of the mold. A larger withdrawal speed can increase throughput, but it also carries risks. As the workpiece moves out of the mold and begins to cool, the solidifying outer shell acts as a thin-walled pressure vessel containing the ferrostatic pressure of the molten core. If v_w is too large, the workpiece will not cool fast enough, and this pressure-retaining "wall" can rupture. The result is a catastrophic defect that forces the workpiece to be scrapped, damages production machinery, and endangers workers. Thus, a safety limit must be imposed on v_w. This limit, shown below, is derived in Appendix I.

$$v_w \leq v_w^{(safe)} := \alpha \frac{K^2 L}{\xi_{safe}^2}$$

where:

- α is a safety factor chosen at the plant operator's discretion
- K is a plant-specific proportionality constant derived from empirical data
- L is the axial length of the mold
- ξ_{safe} is the safe shell thickness (see derivation for more details)

We use the output vector

$$\boldsymbol{y} := \begin{pmatrix} h_M \\ h_T \end{pmatrix}$$

where h_M and h_T are as defined previously. This output is observed through the measurement model

$$\boldsymbol{y} = C\boldsymbol{x}$$

Finally, we explore disturbances that affect system performance and controller design. Owing to the high temperatures, mechanical complexity, and harsh conditions associated with steelmaking, disturbances come in many varieties in continuous casting operations. In practice, these disturbances are usually stochastic in nature. Since stochastic disturbances are beyond the scope of EC402, we consider deterministic approximations to three common disturbances in continuous casting:

$$\boldsymbol{d}(\boldsymbol{t}) := \begin{pmatrix} d_l(t) \\ d_n(t) \\ d_w(t) \end{pmatrix}$$

where:

- $d_l(t)$ is an **upstream flow disturbance** from ladle to tundish
- $d_n(t) \in [0,1]$ is a **mold nozzle flow degradation disturbance**
- $d_w(t)$ is a **mechanical disturbance proxy** in the withdrawal mechanism

Applying mass conservation and fluid dynamics formulae to the above quantities, we have derived the coupled two-dimensional nonlinear state space model

$$\dot{h_M} = \frac{1}{A_M}\left(u_M(1-d_n)A_{nozzle}\sqrt{2g(h_T-h_M)} - (1+d_w)v_w A_M\right)$$

$$\dot{h_T} = \frac{1}{A_T}\left(\overline{Q_{ladle}} - d_l - u_M(1-d_n)A_{nozzle}\sqrt{2g(h_T-h_M)}\right)$$

This model is derived from Bernoulli's principle and conservation laws (obtained by consulting Munson et al., 2013) in Appendix II.

3.2 Definition of Operating Point

Having defined the nonlinear state space model and the relevant system parameters, we now analyze the system's equilibria and define an operating point. By definition, an equilibrium point $(\boldsymbol{x}^\star, \boldsymbol{u}^\star, \boldsymbol{d}^\star)$ satisfies

$$\dot{h_M} = \dot{h_T} = 0$$

In our model, this condition translates into the following system-level flow balance condition (derived in Appendix III):

$$\overline{Q_{ladle}} - d_l = (1+d_w)v_w A_M$$

Our system is underdetermined at equilibrium and therefore admits a continuum of equilibrium points. As a result, linearization requires the selection of a specific operating point.

In this work, we define the operating point by specifying a desired mold height h_M^*, which serves as the primary controlled variable. The corresponding tundish height and input values are then determined from the equilibrium equations outlined in Appendix III. This approach ensures that the linearized model is constructed around a physically meaningful and operationally relevant configuration of the system.

To fully specify the operating point, we assign nominal input values $\overline{u_M}$ and $\overline{v_W}$. The remaining state and disturbance quantities are then determined by enforcing the equilibrium conditions derived above (see Appendix III). In particular, the tundish height $h_T^*$and required ladle flow Q_{ladle}^* are computed so that the steady-state flow balance is satisfied.

Thus, the operating point may be expressed symbolically as

$$\boldsymbol{x}^* = \begin{pmatrix} \overline{h_M} \\ h_T^* \end{pmatrix}$$

$$\boldsymbol{u}^* = \begin{pmatrix} \overline{u_M} \\ \overline{v_W} \end{pmatrix}$$

$$\boldsymbol{d}^* = \begin{pmatrix} d_l^* \\ d_n^* \\ d_w^* \end{pmatrix}$$

3.4 Construction and Analysis of Linearized Model

Linearization involves the computation of three Jacobians from the state space model: a state Jacobian J_A, an input Jacobian J_B, and a disturbance Jacobian J_C. These Jacobians, evaluated at the aforementioned operating point, are presented below.

$$J_A = \begin{pmatrix} \frac{-A_{nozzle}\overline{u_M}g(1-d_n^*)}{A_M\sqrt{2g(h_T^*-\overline{h_M})}} & \frac{A_{nozzle}\overline{u_M}g(1-d_n^*)}{A_M\sqrt{2g(h_T^*-\overline{h_M})}} \\ \frac{-A_{nozzle}\overline{u_M}g(1-d_n^*)}{A_T\sqrt{2g(h_T^*-\overline{h_M})}} & \frac{A_{nozzle}\overline{u_M}g(1-d_n^*)}{A_T\sqrt{2g(h_T^*-\overline{h_M})}} \end{pmatrix}$$

$$J_B = \begin{pmatrix} \frac{A_{nozzle}(1-d_n^*)\sqrt{2g(h_T^*-\overline{h_M})}}{A_M} & (1+d_w^*) \\ \frac{-A_{nozzle}(1-d_n^*)\sqrt{2g(h_T^*-\overline{h_M})}}{A_T} & 0 \end{pmatrix}$$

$$J_C = \begin{pmatrix} 0 & \frac{-u_M A_{nozzle}\sqrt{2g(h_T^*-\overline{h_M})}}{A_M} & -v_W \\ \frac{-1}{A_T} & \frac{u_M A_{nozzle}\sqrt{2g(h_T^*-\overline{h_M})}}{A_T} & 0 \end{pmatrix}$$

We then define deviation coordinates around the operating point:

$$\tilde{\boldsymbol{x}} := \begin{pmatrix} h_M - \overline{h_M} \\ h_T - h_T^* \end{pmatrix}$$

$$\tilde{\boldsymbol{u}} := \begin{pmatrix} u_M - \overline{u_M} \\ v_W - \overline{v_W} \end{pmatrix}$$

$$\tilde{\boldsymbol{d}} := \begin{pmatrix} d_l - d_l^* \\ d_n - d_n^* \\ d_w - d_w^* \end{pmatrix}$$

The linearized model is then constructed as:

$$\dot{\tilde{\boldsymbol{x}}} = J_A\tilde{\boldsymbol{x}} + J_B\tilde{\boldsymbol{u}} + J_C\tilde{\boldsymbol{d}}$$

To verify the feasibility of state feedback and observer design, the linearized model is tested for controllability and observability by computing and applying a rank criterion to the controllability and observability matrices. These are defined respectively as

$$\mathcal{C} := [J_B \;\; J_A J_B]$$

$$\mathcal{O} := \begin{bmatrix} C \\ CJ_A \end{bmatrix}$$

The open-loop stability of the linearized model is then evaluated by computing the eigenvalues of the state Jacobian J_A and checking if J_A is Hurwitz. The above open-loop analysis is performed numerically, and the results are described in *Simulations*.

To ensure physical plausibility in simulation, we enforce a unidirectional flow constraint $h_T \geq h_M$ by modifying the nonlinear dynamics outside the nominal operating region. Since the operating point satisfies $h_T^* > h_M^*$ by construction, this modification does not affect the local linearization.

3.5 State Feedback Controller: Construction and Design Considerations

After the open-loop characteristics of the linearized model are analyzed, two controller designs are evaluated using a suite of performance metrics. The first is a state-feedback controller, designed by performing pole placement on the linearized model closed-loop matrix

$$A_{CL} \coloneqq J_A - J_B K$$

for controller gains K. Our pole placement approach is heavily influenced by operational considerations. Minimizing oscillation in mold height is an important quality control priority for steelmakers. Therefore, an oscillation intensity proxy defined by

$$A_{osc} \coloneqq \sum_k |e_{k+1} - e_k|$$

over simulation timestamps k is used as part of the controller evaluation process. Larger values of this proxy correspond to greater ringing, an operationally undesirable behavior.

In addition to approximating ringing, we analyzed absolute overshoot and maximum absolute error across simulation sample paths as measures of controller performance. Our controller design also focused on minimizing deviations from the operating point. We thus computed metrics including the maximum absolute mold level error, settling time, and steady-state error across simulation sample paths. Since our simulation was conducted in finite time, we could not compute steady-state error as an infinite limit. We instead used the terminal mold height in each simulation sample path as a proxy for the steady-state value and compared that to the setpoint to approximate the steady-state error. Finally, we penalized large deviations from the setpoint by defining a pair of *operating bands* about the setpoint. The *primary operating band* defined as a mold level within $\pm$ 5mm of the specified operating point, and *severe operating band violations* being defined as mold level excursions exceeding 10mm in magnitude from the operating point. The proportion of simulation samples in which each band is violated was a highly weighted consideration in our controller design process.

In addition to quality considerations, safety limits and physical plausibility must also be respected. Thus, all controller simulation sample paths were checked for instances of the mold or tundish levels exceeding container heights (corresponding to a catastrophic molten metal overflow) and for physically implausible negative mold and/or tundish levels.

Alongside the above state feedback controller design and analysis, we explore two extensions to the EC402 course material.

3.6 Extensions: LQR and Luenberger Observer

First, we extend the state feedback framework using the Linear Quadratic Regulator (LQR), which provides an optimal control law that balances regulation performance and control effort. The LQR formulation defines a quadratic cost on state deviation and input magnitude, and yields a feedback law of the form $u = -Kx$, where $K = R^{-1}B^T P$and $P$is the solution to the algebraic Riccati equation (Åström and Murray, 2008). In this work, the weighting matrices are selected to reflect operational priorities, penalizing deviations in mold height more heavily than control effort. The specific weighting design is detailed in **Appendix IV.**

Next, we define the partial observability case

$$C = [1 \quad 0]$$

such that the mold level h_M is observable but the tundish level h_T is not. For this case, we construct a Luenberger observer to estimate the system state from available measurements. The observer augments the system with a correction term based on measurement residuals, yielding estimation error dynamics governed by $\dot{e} = (A - LC)e$. The observer gain L is chosen via pole placement to ensure fast convergence of the estimation error (Antsaklis & Michel, 2006).

These extensions enable the evaluation of controller performance under both full-state and output-feedback configurations.

4. Simulations

4.1 Simulation Pipeline

Simulations were executed using a parameterized experiment pipeline implemented in MATLAB. Rather than manually constructing each case, the code generates simulation parameter grids over operating points, disturbance scenarios, observability cases, controller types, and controller parameter settings. This structure enables systematic and consistent evaluation across linear and nonlinear plants, full-state and partial observability cases, and both state-feedback and LQR controllers under multiple disturbance scenarios. For each run, the pipeline computes performance metrics, checks safety and feasibility constraints, generates plots, and aggregates results into a summary table used for comparative analysis. Plant geometry was fixed as a constant across all simulations.

For the sake of brevity, full exposition of the pipeline architecture and simulation results is omitted from the report. Instead, the code is made available via a public GitHub repository. We now present a selection of targeted experiments aimed at evaluating controller performance along operationally relevant dimensions.

4.2 Experiment 1: What pole locations yield the best operating performance for the state feedback controller?

Our scope of experimentation commenced with a sweep of candidate pole pairs for the state feedback controller. All subsequent experiments used the poles placed in this sweep. To perform the sweep, we defined a 2-dimensional geometric grid of pole pairs with the form

$$(p_1, p_2) \coloneqq (-\alpha, -\rho\alpha)$$

The parameter α controls the "aggressiveness" of the dominant pole (and thereby modulates convergence speed), while the parameter ρ governs the separation between the two poles. We evaluated the candidate poles within this grid along two dimensions of particular operational interest: control energy and maximum mold level deviation. Comparing performance along these two dimensions elucidates tradeoffs between mold level regulation and control effort, each of which carries economic consequences. We then construct the Pareto frontier shown in Figure 4.2.1.

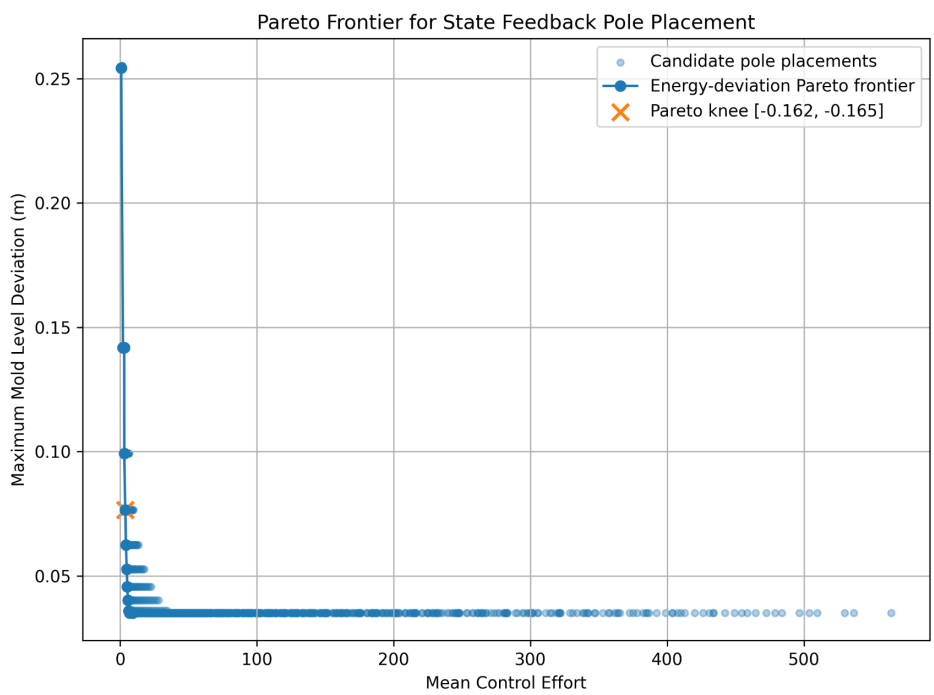

Figure 4.2.1: Pareto Frontier for State Feedback Pole Placement

A **Pareto knee** is identified as the point of maximum curvature on the energy-deviation tradeoff curve (as approximated from the discrete Pareto frontier). This is the point at which marginal improvements in deviation begin to require disproportionately large increases in control effort. We thus apply a heuristic pole selection rule, taking as our chosen operating configuration the poles at the Pareto knee. This selection rule reflects a diminishing-returns principle: beyond the Pareto knee, improvements in regulation quality are outweighed by rapidly increasing control effort, making such designs unattractive in practice.

4.3 Experiment 2: Does LQR provide a meaningful operational advantage vis-à-vis state feedback?

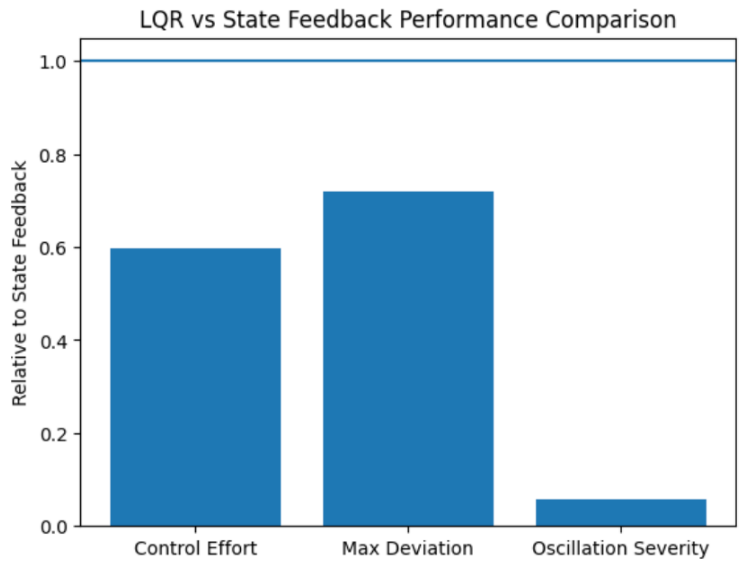

Figure 4.2.2: LQR vs State Feedback Performance Comparison

The LQR controller is evaluated against the selected pole-placement controller under identical disturbance scenarios and full-state observability. Figure 4.3.1 shows performance normalized to the state-feedback baseline, where values below one indicate improvement. The LQR controller significantly reduces control effort (≈40%), maximum mold-level deviation (≈28%), and oscillation severity (≈95%) relative to pole placement. These results indicate that LQR constitutes a strictly superior operating configuration in this setting, achieving both improved regulation performance and reduced actuation effort. This reflects the advantage of directly optimizing a quadratic cost that captures operational objectives, rather than indirectly shaping system dynamics via pole placement.

4.4 Experiment 3: How do the controllers perform under disturbances and nonlinear dynamics?

Disturbance rejection performance was evaluated across a broad set of disturbance scenarios. An indicative example, in which a transient pulse disturbance is applied to the withdrawal speed to induce an abrupt outflow imbalance, is shown here. Figure 4.4.1 shows the nonlinear mold level trajectory under this disturbance, while Figure 4.4.2 shows the corresponding control inputs.

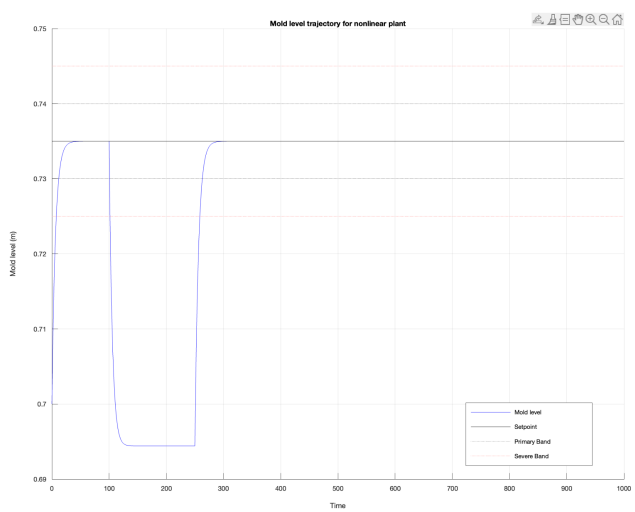

Figure 4.4.1: Mold level trajectory for nonlinear plant under pulse withdrawal disturbance

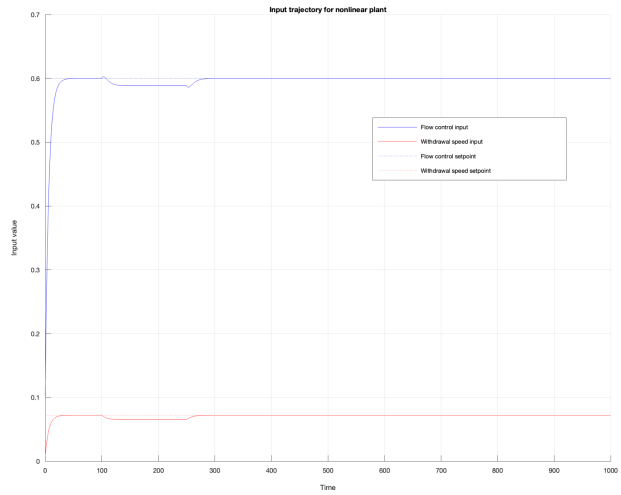

Figure 4.4.2: Control inputs for nonlinear plant under pulse withdrawal disturbance

The controller maintains closed-loop stability and returns the system to the operating point following the disturbance. However, the transient response reveals important limitations. In particular, the mold level exhibits significant excursions below the setpoint, temporarily violating the primary operating band. While recovery is ultimately achieved, the magnitude and duration of these deviations would be undesirable in a real casting process.

The input trajectory remains relatively smooth and low-amplitude, indicating that the controller does not increase actuation sufficiently to counteract the disturbance-induced outflow imbalance. This suggests a tradeoff inherent in the controller design: prioritizing moderate control effort leads to acceptable stability, but at the cost of degraded disturbance rejection performance in more severe scenarios.

Larger and longer-lived excursions were observed for persistent disturbances (constant and step disturbances), indicating that some of the observed recovery is attributable to the cessation of the disturbance rather than active rejection by the controller. The general conclusion observed across trials is that disturbance rejection is acceptable for mild disturbances, but deteriorates under more aggressive perturbations, indicating that additional tuning or alternative control strategies may be required for industrial deployment.

4.5 Experiment 4: How does partial observability impact system performance?

The impact of partial observability was evaluated by comparing full-state feedback against output feedback using a Luenberger observer with mold-level measurements only. Figures 4.5.1 and 4.5.2 summarize the resulting performance differences.

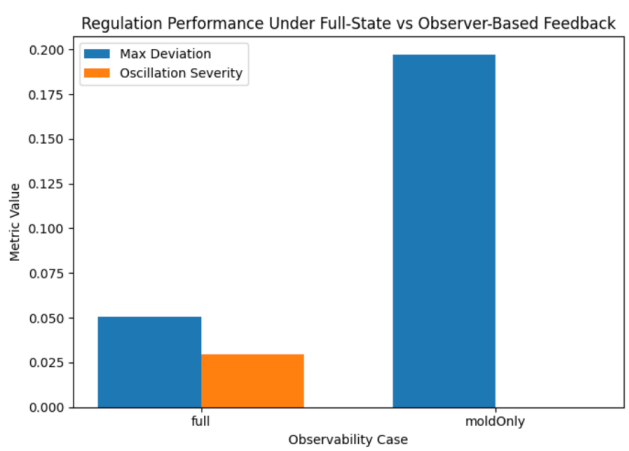

Figure 4.5.1: Regulation Performance Under Full-State vs Observer-Based Feedback

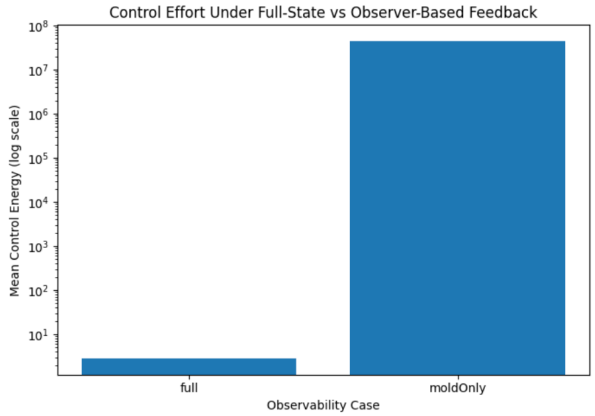

Figure 4.5.2: Control Effort Under Full-State vs Observer-Based Feedback

Regulation performance remains broadly comparable between the two cases: maximum mold level deviation and oscillation severity show only minor differences under partial observability. This indicates that the observer is sufficient to maintain closed-loop stability and approximate state regulation.

However, a substantial increase in control effort is observed when the observer is used. In particular, control energy increases by several orders of magnitude under partial observability. This behavior arises because the controller acts on estimated rather than true states, leading to inefficient and overly aggressive actuation to compensate for estimation error.

Overall, partial observability introduces a significant tradeoff: acceptable regulation performance can be maintained, but at the cost of dramatically increased control effort. Such behavior may be impractical in real industrial settings, where actuator limitations and energy costs are critical considerations.

5. Tasks and Deliverables

The list of tasks presented below mirrors the project proposal. <u>All proposed tasks have been completed fully.</u>

a. Derive state space model from system physics (**100% completion** – presented in Appendix II)
b. Analyze equilibria and linearize about operating point (**100% completion** – presented in 3.1 – 3.2)
c. Evaluate stability of linearized system (**100% completion** – calculated automatically by our MATLAB pipeline and not explicitly mentioned in the report)
d. Evaluate controllability and observability of linearized system (**100% completion** – matrices $\mathcal{C}$ and $\mathcal{O}$ are defined analytically in the report, and are computed and checked for full rank in each run of the simulation pipeline)
e. Design state feedback controller and perform pole placement (**100% completion** – see and 3.5, 4.2)
f. Design observer (**100% completion** – see 3.6 and 4.5)
g. Design LQR control laws (**100% completion** – see 3.6, 4.3, and Appendix IV)
h. Simulate system and controller in MATLAB to validate/extend analytical results (**100% completion**)

6. Conclusions

This project investigated the regulation of mold height in a continuous casting system using state-space control methods under realistic constraints, including disturbances, nonlinear dynamics, and partial observability. The results yield several insights of direct relevance to industrial control design.

First, controller tuning must explicitly account for tradeoffs between regulation performance and control effort. The pole-placement sweep demonstrated that aggressive pole configurations can reduce mold level deviation, but only at the cost of rapidly increasing control energy. The identification of a Pareto knee provides a practical heuristic for selecting operating points that balance quality and operational cost, reflecting the diminishing returns observed in real industrial systems.

Second, the Linear Quadratic Regulator (LQR) framework provides a clear advantage over heuristic pole placement. By directly encoding operational objectives into a cost function, LQR produced controllers that simultaneously reduced mold level deviation, oscillation severity, and control effort. This suggests that optimization-based control design is better

aligned with industrial requirements than manual pole tuning, particularly in systems where multiple competing objectives must be balanced.

Third, disturbance rejection remains a critical challenge. While the controllers maintain closed-loop stability under transient disturbances, significant excursions from the operating point were observed, particularly under pulse and persistent disturbances. These deviations violate practical operating constraints and would likely lead to defects or safety risks in a real casting process. Importantly, recovery in some cases is driven more by the cessation of the disturbance than by active rejection, indicating that the controllers are not inherently robust to sustained perturbations. This highlights the need for more advanced disturbance-handling strategies in industrial deployments.

Fourth, partial observability introduces a severe and often underappreciated cost. Although the Luenberger observer enables stable output-feedback control with only mold-level measurements, it does so at the expense of dramatically increased control effort. The resulting actuation levels are likely impractical in real systems due to energy costs and actuator limitations. This finding reinforces the importance of high-quality sensing and suggests that investment in improved measurement infrastructure may yield greater returns than purely algorithmic enhancements.

Taken together, these results emphasize that effective control design in continuous casting is not solely a matter of achieving stability, but of navigating a multi-dimensional tradeoff between regulation quality, robustness, actuation cost, and sensing limitations. Controllers that perform well in nominal conditions may fail to meet practical requirements when disturbances and observability constraints are introduced.

Several extensions would further improve the realism and applicability of this work. Incorporating stochastic disturbance models would better capture the variability present in industrial environments. More detailed actuator models could account for physical limitations and nonlinearities in flow control hardware. Finally, integrating an explicit economic cost model would enable direct evaluation of control policies in terms of production efficiency and defect risk, providing a stronger bridge between control design and operational decision-making.

Generative AI Disclosure

AI tools (OpenAI ChatGPT and Google Gemini) were used to assist in visualization of simulation outputs (e.g., generating Pareto frontier plots from computed metrics). They were also employed for assistance in understanding industry-specific technical background and literature on steelmaking. All models, experiments, analytical work, selection criteria, and interpretation of results were the exclusive purview of the authors.

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Appendix I: Derivation of Withdrawal Speed Safety Limit

Let ξ be the **solidification front**, the position of the solid-liquid interface within the workpiece. This quantity varies as the workpiece moves out of the mold.

To derive the safety limit, we consider ξ at the point where it is smallest (the point where the workpiece exits the mold). If v_w is sufficient to avoid structural failure at this point, it should also be sufficient for all other points along the workpiece's trajectory.

Per Thomas (2002), the solidification front may be modelled as

$$\xi = K\sqrt{\frac{z}{v_w}}$$

Where z is the distance travelled by the workpiece and K is an empirically determined, plant-specific proportionality constant. Letting the axial dimension of the mold equal L and measuring the solidification front at the exit of the mold, $z = L$. For safety, the solidification front must exceed some minimum threshold ξ_{safe}. Thus, we require:

$$\xi_{safe} \leq K\sqrt{\frac{L}{v_w}}$$

Which leads to the withdrawal speed safety requirement

$$v_w \leq v_w^{(safe)} := \frac{K^2 L}{\xi_{safe}^2}$$

We then apply a safety factor α to ensure that the operating point is kept a safe distance from the limit under stochastic operating perturbations (whose nature is beyond the scope of the EC402 course and of our analysis). This leads to the final safe withdrawal speed:

$$\boldsymbol{v_w \leq v_w^{(safe)} := \alpha \frac{K^2 L}{\xi_{safe}^2}}$$

Appendix II: Derivation of the Full State Space Model

We derive the nonlinear state space model from conservation of mass and Bernoulli's principle. The relevant physical laws were obtained from Munson et al. (2013).The two state variables are the mold level $h_M(t)$ and the tundish level $h_T(t)$. We assume that the relevant cross-sectional areas are constant, so changes in molten steel volume may be written in terms of height changes.

For any tank volume with cross-sectional area A, conservation of mass gives for the general flow Q.

$$\frac{d}{dt}(Ah) = Q_{\text{in}} - Q_{\text{out}}.$$

Since A is constant, this becomes

$$\dot{h} = \frac{1}{A}(Q_{\text{in}} - Q_{\text{out}}).$$

We now apply this relation separately to the mold and the tundish.

For the mold, inflow is the flow from the tundish through the submerged entry nozzle, while outflow is caused by withdrawal of the steel strand. The flow into the mold is regulated by the actuator $u_M(t) \in [0,1]$, which represents the slide-gate or stopper-rod opening. Thus,

$$\dot{h}_M = \frac{1}{A_M}\left(u_M Q_{\text{in}}^{(M)} - Q_{\text{out}}^{(M)}\right).$$

To model the nozzle inflow, we apply Bernoulli's principle between the tundish free surface and the nozzle exit into the mold. We assume both free surfaces are exposed to approximately atmospheric pressure, and the velocity at the tundish free surface is negligible relative to the nozzle velocity. Therefore,

$$P_{\text{atm}} + \rho g h_T = P_{\text{atm}} + \rho g h_M + \frac{1}{2}\rho v_{\text{nozzle}}^2.$$

Canceling atmospheric pressure and dividing by ρgives

$$g h_T = g h_M + \frac{1}{2} v_{\text{nozzle}}^2,$$

so

$$v_{\text{nozzle}} = \sqrt{2g(h_T - h_M)}.$$

Since flow rate is velocity times area,

$$Q_{\text{in}}^{(M)} = A_{\text{nozzle}}\sqrt{2g(h_T - h_M)}.$$

The mold outflow is the withdrawal of the strand through the mold cross-sectional area:

$$Q_{\text{out}}^{(M)} = A_M v_W.$$

Substituting into the mold height balance gives

$$\dot{h}_M = \frac{1}{A_M}\left(u_M A_{\text{nozzle}}\sqrt{2g(h_T - h_M)} - A_M v_W\right).$$

We then incorporate the disturbance terms used in the project. The nozzle degradation disturbance $d_n(t) \in [0,1]$reduces the effective nozzle flow area by the factor $(1-d_n)$, while the withdrawal disturbance $d_w(t)$modifies the effective withdrawal speed by the factor $(1+d_w)$. Thus,

$$\boxed{\dot{h}_M = \frac{1}{A_M}\left(u_M(1-d_n)A_{\text{nozzle}}\sqrt{2g(h_T - h_M)} - (1+d_w)v_W A_M\right)}$$

For the tundish, the inflow is the ladle flow Q_{ladle}, while the outflow is the same tundish-to-mold flow entering the mold. Conservation of mass gives

$$\dot{h}_T = \frac{1}{A_T}\left(Q_{\text{in}}^{(T)} - Q_{\text{out}}^{(T)}\right).$$

The tundish inflow is

$$Q_{\text{in}}^{(T)} = Q_{\text{ladle}},$$

and conservation of mass requires the tundish outflow to equal the mold inflow:

$$Q_{\text{out}}^{(T)} = u_M A_{\text{nozzle}}\sqrt{2g(h_T - h_M)}.$$

Including the upstream ladle disturbance $d_l(t)$, which reduces the effective flow from ladle to tundish, and the nozzle degradation disturbance $d_n(t)$, the tundish dynamics become

$$\boxed{\dot{h}_T = \frac{1}{A_T}\left(Q_{\text{ladle}} - d_l - u_M(1-d_n)A_{\text{nozzle}}\sqrt{2g(h_T - h_M)}\right)}$$

Combining these equations yields the nonlinear state space model used in the project:

$$\boxed{\begin{aligned} \dot{h}_M &= \frac{1}{A_M}\left(u_M(1-d_n)A_{\text{nozzle}}\sqrt{2g(h_T - h_M)} - (1+d_w)v_W A_M\right), \\ \dot{h}_T &= \frac{1}{A_T}\left(Q_{\text{ladle}} - d_l - u_M(1-d_n)A_{\text{nozzle}}\sqrt{2g(h_T - h_M)}\right). \end{aligned}}$$

This defines the nonlinear dynamics

$$\dot{x} = f(x, u, d),$$

where

$$x = \begin{bmatrix} h_M \\ h_T \end{bmatrix}, u = \begin{bmatrix} u_M \\ v_W \end{bmatrix}, d = \begin{bmatrix} d_l \\ d_n \\ d_w \end{bmatrix}.$$

The term $\sqrt{2g(h_T - h_M)}$is physically meaningful only when $h_T \geq h_M$, corresponding to gravity-driven flow from the tundish into the mold. In simulation, the nonlinear model is therefore evaluated with a non-reverse-flow extension outside the nominal operating region. Since the operating point used for linearization satisfies $h_T^* > h_M^*$, this extension does not alter the local Jacobian used for controller design.

Appendix III: Equilibrium Analysis for Nonlinear State Space Model

We now characterize the equilibrium conditions for the nonlinear model derived in Appendix II. An equilibrium $(x^*, u^*)$is defined by the condition $\dot{x} = 0$, i.e., the system must satisfy a steady flow balance with no accumulation in either the mold or the tundish.

Recalling the dynamics,

$$\dot{h}_M = \frac{1}{A_M}\left(u_M(1-d_n)A_{\text{nozzle}}\sqrt{2g(h_T - h_M)} - (1+d_w)v_W A_M\right),$$
$$\dot{h}_T = \frac{1}{A_T}\left(Q_{\text{ladle}} - d_l - u_M(1-d_n)A_{\text{nozzle}}\sqrt{2g(h_T - h_M)}\right),$$

we impose $\dot{h}_M = \dot{h}_T = 0$. This yields

$$0 = u_M^*(1-d_n)A_{\text{nozzle}}\sqrt{2g(h_T^* - h_M^*)} - (1+d_w)v_W^* A_M,$$
$$0 = Q_{\text{ladle}} - d_l - u_M^*(1-d_n)A_{\text{nozzle}}\sqrt{2g(h_T^* - h_M^*)}.$$

At this point the structure becomes clear. Both equations contain the same nonlinear flow term, so it is convenient to define

$$Q_{T\to M}^* = u_M^*(1-d_n)A_{\text{nozzle}}\sqrt{2g(h_T^* - h_M^*)}.$$

Substituting, the equilibrium conditions reduce to

$$Q_{T\to M}^* = (1+d_w)v_W^* A_M, Q_{T\to M}^* = Q_{\text{ladle}} - d_l.$$

Combining these immediately gives a single global balance condition,

$$Q_{\text{ladle}} - d_l = (1+d_w)v_W^* A_M.$$

This is the only hard constraint imposed by equilibrium: inflow from the ladle must equal outflow through withdrawal. Solving for the withdrawal speed,

$$v_W^* = \frac{Q_{\text{ladle}} - d_l}{A_M(1+d_w)}.$$

The more interesting point is what *isn't* fixed. The internal variables h_M^*, h_T^*, and u_M^*do not have a unique solution. Instead, they must satisfy

$$u_M^*(1-d_n)A_{\text{nozzle}}\sqrt{2g(h_T^*-h_M^*)} = Q_{\text{ladle}} - d_l,$$

which defines a continuum of valid equilibrium configurations. Increasing the valve opening u_M^*can be offset by reducing the hydraulic head $h_T^* - h_M^*$, and vice versa. From a purely mathematical standpoint, the system admits infinitely many equilibria.

This is slightly nonstandard from a controls perspective. The equilibrium is not uniquely determined by the system itself but instead depends on how the inputs are chosen. In practice, this is resolved by selecting a desired operating point — typically a target mold level h_M^*— and then backing out the corresponding tundish level and valve position required to sustain the associated flow.

The key takeaway is that equilibrium is governed by a simple global flow balance, but the internal state configuration achieving that balance is not unique. This has direct implications for control design: we are not stabilizing to a single natural equilibrium but rather regulating the system about a chosen operating condition defined by process requirements.

Appendix IV: Analysis of Linear Quadratic Regulator (LQR) Formulation

We consider the standard infinite-horizon quadratic cost functional

$$J = \int_0^\infty (x^T Q x + u^T R u)\ dt,$$

where $x \in \mathbb{R}^2$and $u \in \mathbb{R}^2$in our setting. The objective is to construct a state feedback law $u = -Kx$that minimizes this cost for the linearized dynamics about the chosen operating point.

We do not attempt to solve this problem analytically from first principles. Instead, following standard practice (e.g., Åström and Murray), we specify the weighting matrices $Q$and Rdirectly based on physical considerations, and then compute the resulting optimal controller via the algebraic Riccati equation.

The key design choice lies in how $Q$and R are constructed.

Our approach is to weight each state and input according to its allowable deviation in operation. Let $\Delta h_M$and Δh_Tdenote characteristic tolerances for mold level and tundish level, respectively, and let Δu_M, Δv_Wdenote characteristic actuation scales for the valve position and withdrawal speed. We then define

$$Q = \begin{bmatrix} \frac{1}{(\Delta h_M)^2} & 0 \\ 0 & \frac{1}{(\Delta h_T)^2} \end{bmatrix}, R = \begin{bmatrix} \frac{1}{(\Delta u_M)^2} & 0 \\ 0 & \frac{1}{(\Delta v_W)^2} \end{bmatrix}.$$

This construction penalizes deviations in proportion to the inverse square of their acceptable magnitudes. The effect is that a deviation equal to its tolerance contributes order-one cost, while larger deviations are rapidly penalized. In other words, the cost functional is implicitly normalized by operational limits.

This choice is directly motivated by the physical constraints of the casting process. Mold level deviations beyond a few millimeters are known to be highly undesirable due to risks such as air entrainment or surface defects, while tundish level deviations are comparatively less critical. Similarly, aggressive actuation — particularly in withdrawal speed — is limited by mechanical and process safety considerations. The weighting scheme reflects these asymmetries without introducing additional tuning parameters.

Given these definitions, the optimal state feedback gain is obtained by solving the continuous-time algebraic Riccati equation

$$A^T P + PA - PBR^{-1}B^T P + Q = 0,$$

for the positive definite matrix $P \in \mathbb{R}^{2\times 2}$. The resulting control law is

$$u = -Kx, K = R^{-1}B^T P.$$

This yields the LQR controller used in the simulation studies.

One practical detail is that the withdrawal speed is subject to a safety constraint of the form

$$v_W \leq v_W^{\max},$$

derived in Appendix I. While this constraint is not explicitly enforced within the LQR formulation, it informs the choice of $\Delta v_W$and therefore the corresponding entry in R, ensuring that the controller does not incentivize unrealistically large actuation.

Overall, the LQR design provides a principled alternative to pole placement, replacing heuristic eigenvalue selection with a cost-based formulation grounded in the physical tolerances of the system. The resulting controller trades off regulation performance against control effort in a way that is directly interpretable in terms of operational limits, which is consistent with the objectives of the process.